Practice assignment for Lesson 2.3.1

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Practice Assignment • 10 min

English

Your grade: 100%

Your latest: **100%** • Your highest: **100%**

To pass you need at least 80%. We keep your highest score.

Next item

1. Suppose that

1 / 1 point

$\tilde{z}_k = \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix}$ and $\Sigma_{\tilde{z}_k} = \begin{bmatrix} 2 & 0.1 \\ 0.1 & 2 \end{bmatrix}$.

What is the "normalized estimation error squared (NEES)" e_k^2? Enter your answer with two digits to the right of the decimal point. You may use Octave or similar tool to assist with the calculations.

0.15

Correct
Yes. This is the correct answer.

2. Using Octave or MATLAB, compute χ_U^2 for $\alpha = 0.02$ and $m = 1$. Enter your answer with one digit to the right of the decimal point.

1 / 1 point

5.4

Correct
Yes. This is the correct answer.

3. Using the table provided in the notes for this lesson, compute χ_U^2 for $\alpha = 0.01$ and $m = 3$. Enter the value exactly as written in the table.

1 / 1 point

11.345

Correct
Yes, this is the correct answer.